Memory X-Ray

Does a language model cite the evidence it actually uses?

We removed evidence in every combination to see what drives a model's answer, then checked its citations.

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EXPLORE THE PROJECT

Citations can look right without being faithful

AI assistants increasingly show their sources, but a source can be plausible without being faithful [4]:

PLAUSIBLE
The cited evidence **supports** the answer.

FAITHFUL
The cited evidence actually **drove** the answer.

Self-reports need not be faithful [3], so we measure influence from behavior.

RESEARCH QUESTION

When a model says which evidence it relied on, does that match the evidence that actually drives its answer?

Remove evidence, watch the answer

WHAT THE MODEL DOES

1. Give it a question and 4 evidence records (E1–E4).
2. Score its answer on all 16 subsets × 3 orders [2].
3. Credit each record: Shapley value [1] vs leave-one-out.

WHAT THE MODEL SAYS

4. Show it the answer; ask which records it relied on.
5. Compare its citation with same-size alternatives.

Phi-3-mini · 3 designed controls + 3 realistic economic cases · protocol frozen before the run.

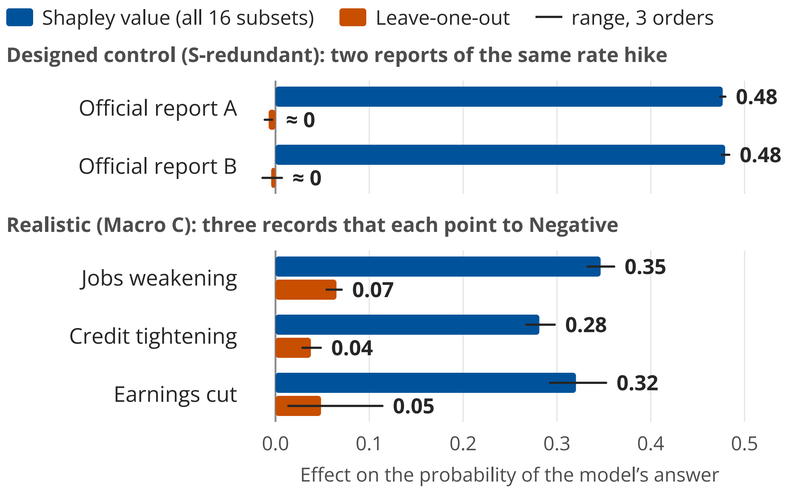
THE MEASUREMENT CHECKS OUT

6 / 6 scenarios pass all 9 infrastructure checks	**2.2×10^{-16}** max. gap vs an independent re-analysis	**2 / 3** controls recover their designed structure

FINDING 1 PRE-REGISTERED

One-at-a-time removal misses backup evidence

If either of two records settles the answer alone, leave-one-out scores both **≈ 0**; Shapley credits each **≈ 0.48**.

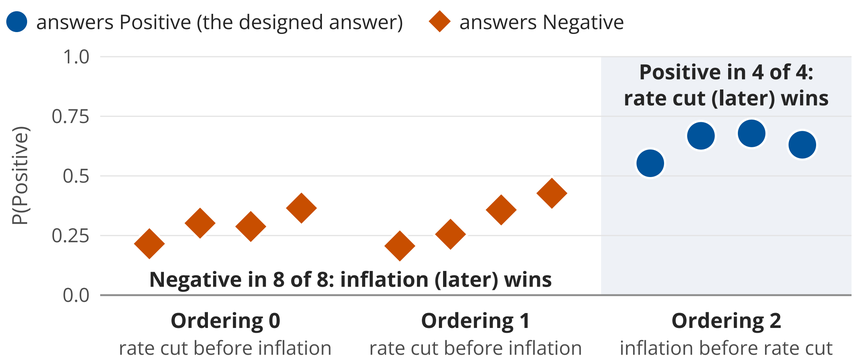


Bars: mean of 3 evidence orders; lines: range (not confidence intervals).

FINDING 2 PRE-REGISTERED CHECKS DESCRIPTIVE COUNT

In our conflict test, order beat the stated rule

The rule said a rate cut overrides an inflation report. In **12 of 12** prompts the later record won; the control failed its checks.

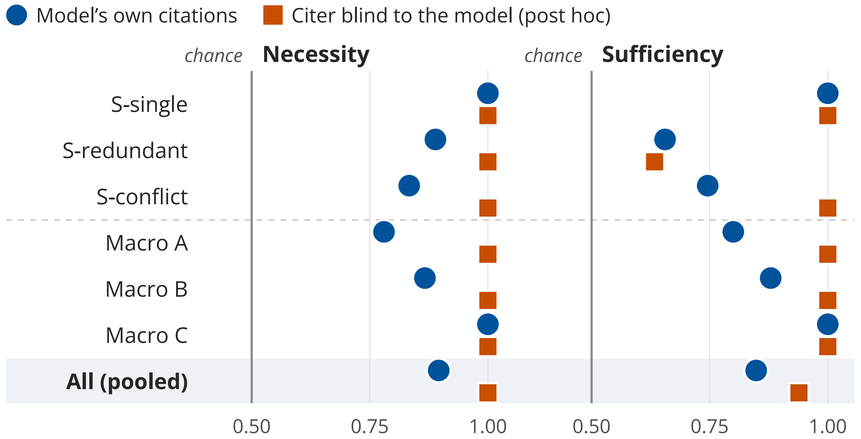


Caveat: one ordering contrast, confounded with position; not a proven recency effect.

FINDINGS 3 & 4

Citations look right, but so does a citer that never sees the model

- Cited sets beat same-size alternatives: **0.90** necessity, **0.85** sufficiency. PRE-REGISTERED In S-conflict, only 2 of 8 replies cited the rule-correct E1 alone. DESCRIPTIVE
- A citer built from our design and rubric, blind to the model, does as well or better in **17 of 18** comparisons (10 higher, 7 tied; not like-for-like). POST HOC
- In 3 of 18 scenario-orderings the label shown was not the model's own; its citations still beat chance, though lower on average. POST HOC



Beat fraction: share of same-size record sets the citation beats. Pooled: 0.90 / 0.85 vs 1.00 / 0.94.

What this means

1. All-subset evidence removal was reproducible here and caught redundancy that leave-one-out misses.
2. Citation agreement is not proof of faithfulness, or of unfaithfulness.
3. Audits should separate faithful from plausible citers, control evidence order and report probability shifts.

Limits: one small model and prompt; six synthetic scenarios; AI-checked rubric; behavioral (not internal) attribution; no significance tests; 23% of sampled replies unparseable. **Next:** cases where supporting and driving evidence differ; more models.

REFERENCES *Full references available on GitHub.*

[1] Shapley (1953). A value for *n*-person games. *Contributions to the Theory of Games II*.
[2] Cohen-Wang et al. (2024). ContextCite: Attributing model generation to context. *NeurIPS*.
[3] Turpin et al. (2023). Language models don't always say what they think. *NeurIPS*.
[4] Wallat et al. (2025). Correctness is not faithfulness in retrieval augmented generation attributions. *ICTIR*.

AI USE AND REPRODUCIBILITY

Rubric scores were drafted with an AI assistant and validated by two AI models, not human annotators. Code and poster drafting were assisted by Claude (Anthropic) and reviewed by the author. All numbers regenerate on a CPU via `scripts/verify_release.py`.